

STAGE 1 — Data Structures

Before: Narratives and sentences were stored as plain dictionaries, like {"id": "P001", "text": "..."} . Simple but unstructured.

Prompt: "Add dataclasses"

After: Dedicated dataclass objects like SentenceAgency and NarrativeSentiment with named fields — more organised and harder to accidentally break.

STAGE 2 — Progress Reporting

Before: Used basic print() statements like print("Loading corpus...") to show what the script was doing.

Prompt: "Replace print statements with the logging module"

After: Uses logging.basicConfig() and logger.info(), which adds timestamps, severity levels, and can be turned on/off without editing the code.

STAGE 3 — Function Signatures

Before: Functions had no indication of what they expected or returned, for example def analyze_agency(nlp, narratives).

Prompt: "Add type hints to all function signatures"

After: Functions clearly declare their inputs and outputs, for example def analyze_agency(nlp, narratives: list[dict]) -> list[NarrativeAgency].

STAGE 4 — Visualisations

Before: 4 basic figures — a bar chart, box plot, scatter plot, and heatmap.

Prompt: "Expand from 4 to 7 figures"

After: 7 figures, adding affective density by author type and voice, negative affect ratio with error bars, and per-narrative timeline charts showing how agency and sentiment shift sentence by sentence.

STAGE 5 — Statistical Depth

Before: Only ran Mann-Whitney and Spearman tests, jumping straight to inferential statistics.

Prompt: "Add a descriptive statistics function"

After: First computes mean, median, standard deviation, and IQR per group before running significance tests — a more complete and academically standard approach.

STAGE 6 — Frequency Distributions

Before: Some basic groupby counts were computed inline but not saved or clearly reported.

Prompt: "Add a frequency distributions function"

After: A dedicated `compute_frequency_distributions()` function produces cross-tabulated counts and means, saved as separate CSV files and included in the console report.

Source: Claude.ai